

Loan Approval Prediction Using Machine Learning

1. Objective

The objective of this project is to develop a machine learning model that predicts whether a loan application will be approved based on applicant information such as income, education, employment status, credit history, and loan amount. The project aims to assist financial institutions in making faster and more accurate loan approval decisions.

2. Dataset Overview

The dataset was obtained from Kaggle and contains historical loan application records.

```
df = pd.read_csv(next(iter(uploaded)))
df.head()
```

...

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	

Next steps: [Generate code with df](#) [New interactive sheet](#)

Key Features

- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area
- Loan_Status

The target variable indicates whether the loan was approved or rejected.

3. Data Preprocessing

Several preprocessing techniques were applied to improve data quality and model performance.

Missing Value Treatment

```
df.isnull().sum()
```

	0
address	17
name	16
online_order	19
book_table	58
rate	7838
votes	78
phone	1296
location	126
rest_type	338
dish_liked	28225
cuisines	203
approx_cost(for two people)	521
listed_in(type)	4610

dtype: int64

Missing values in categorical columns were replaced with the mode, while numerical columns were filled using the median.

Categorical Encoding

Categorical variables were converted into numerical format using Label Encoding.

Feature Scaling

StandardScaler was used to normalize numerical features before model training.

Handling Class Imbalance

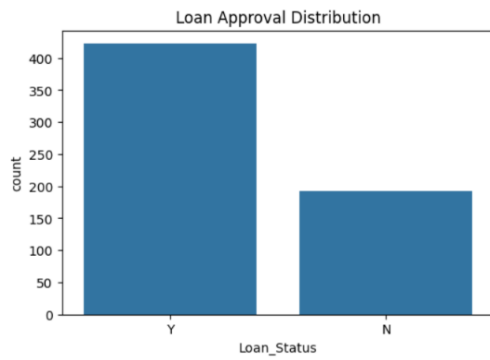
SMOTE (Synthetic Minority Oversampling Technique) was applied to balance the class distribution and improve model learning.

4. Exploratory Data Analysis

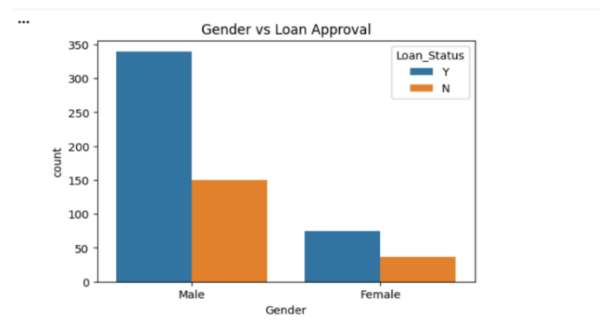
The dataset was analyzed to identify trends and relationships among variables.

Visualizations

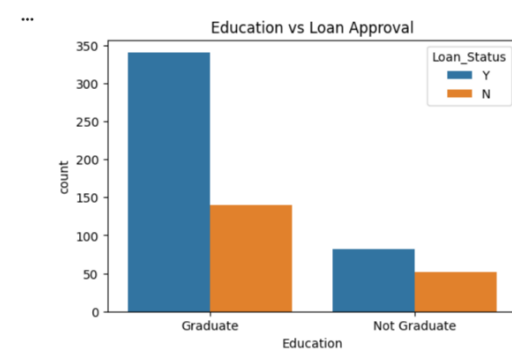
Loan Approval Distribution



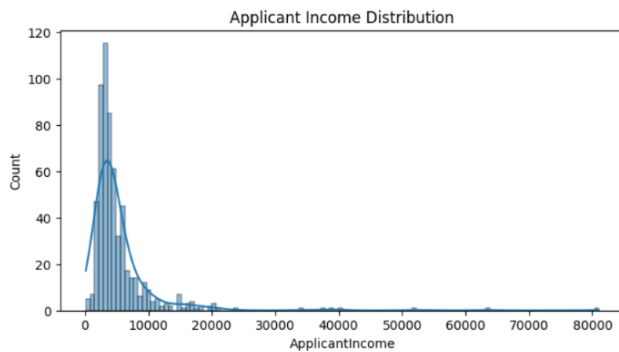
Gender vs Loan Approval



Education vs Loan Approval



Applicant Income Distribution



Class Distribution Before and After SMOTE

```
*** Before SMOTE
Loan_Status
1    337
0    154
Name: count, dtype: int64

After SMOTE
Loan_Status
1    337
0    337
Name: count, dtype: int64
```

5. Machine Learning Models

Three machine learning models were trained and evaluated.

Logistic Regression

A baseline classification model used for binary classification tasks.

Random Forest Classifier

An ensemble learning method that combines multiple decision trees for improved prediction accuracy.

XGBoost Classifier

A gradient boosting algorithm known for its high performance on structured datasets.

6. Model Evaluation

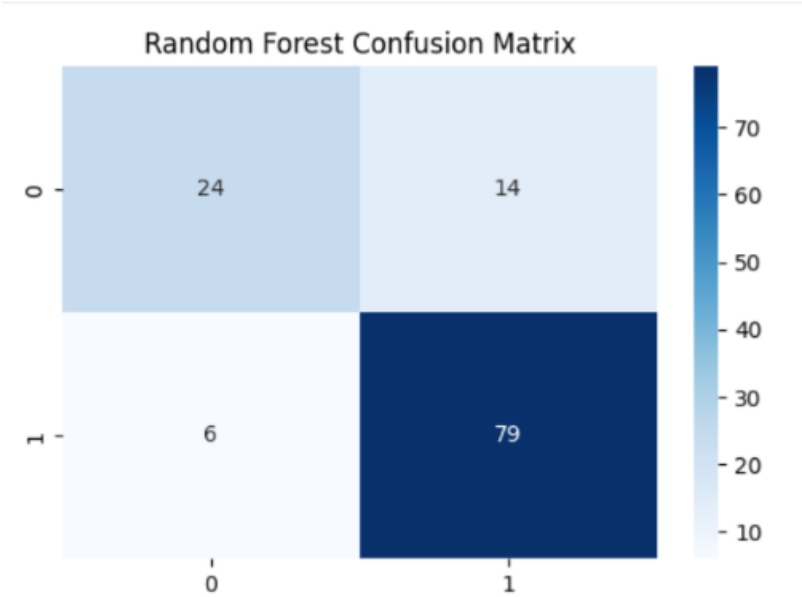
The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

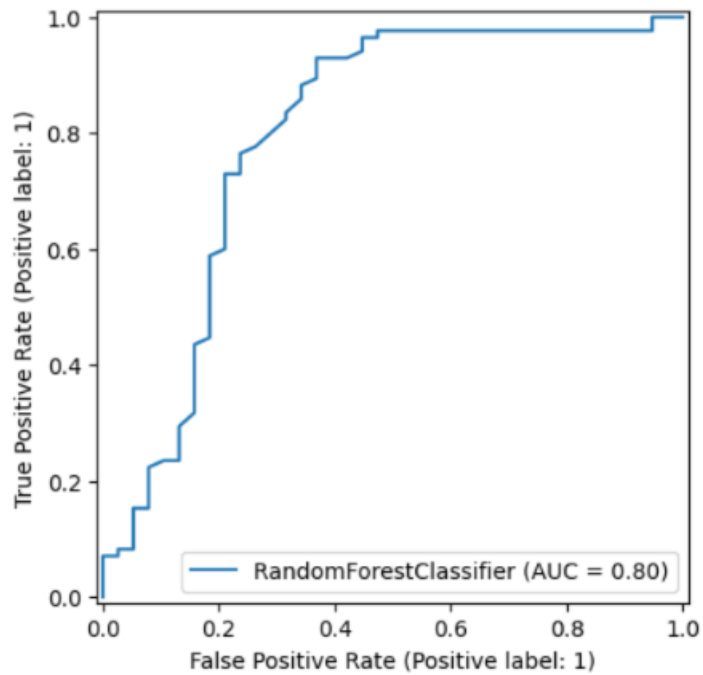
Model Comparison Table

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.821138	0.838710	0.917647	0.876404	0.840557
Random Forest	0.837398	0.849462	0.929412	0.887640	0.796904
XGBoost	0.796748	0.857143	0.847059	0.852071	0.773375

Confusion Matrix



ROC Curve

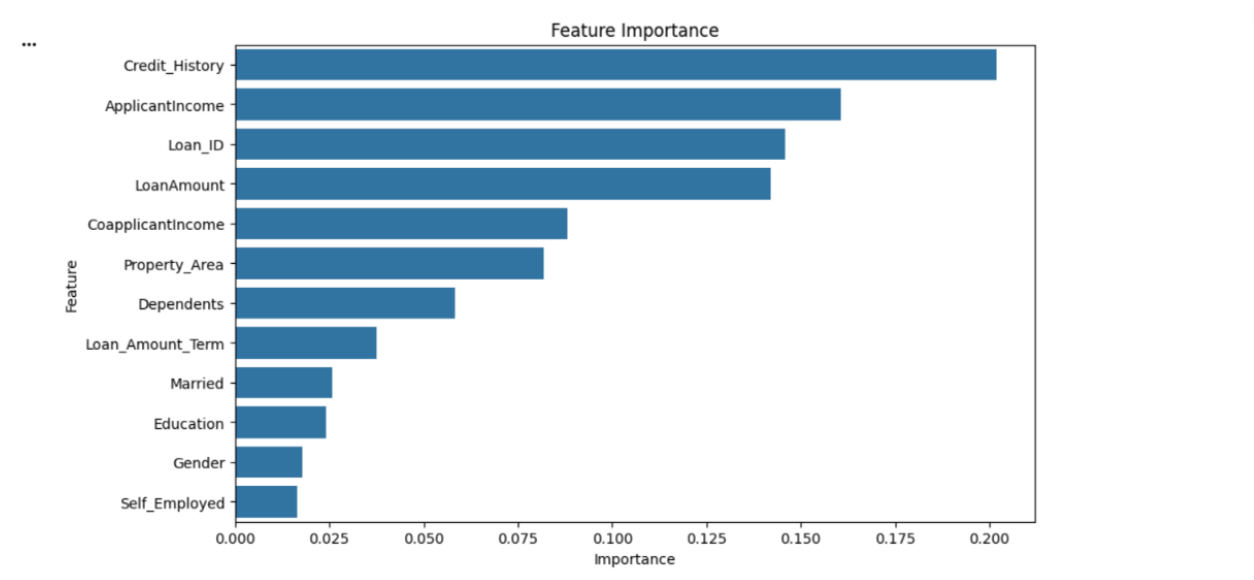


7. Feature Importance Analysis

Feature importance analysis was performed using the Random Forest model to identify the most influential variables affecting loan approval decisions.

Important Features

- Credit History
- Applicant Income
- Loan Amount
- Education
- Property Area



8. Key Findings

1. Credit history was the most significant factor influencing loan approval.
 2. Applicants with stable and higher incomes had greater chances of approval.
 3. Education level showed a positive impact on approval rates.
 4. Class balancing using SMOTE improved the model's ability to identify minority cases.
 5. Ensemble models outperformed the baseline Logistic Regression model.
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9. Business Recommendations

1. Prioritize credit history during loan evaluation.
 2. Implement automated loan screening using the trained model.
 3. Use probability-based approval thresholds to reduce risk.
 4. Conduct additional verification for borderline applications.
 5. Periodically retrain the model with updated loan records to maintain accuracy.
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10. Conclusion

This project successfully developed a machine learning-based loan approval prediction system using historical applicant data. Data preprocessing, feature engineering, class balancing, and model evaluation techniques were applied to improve predictive performance. Among the evaluated models, the best-performing model demonstrated strong capability in predicting loan approvals and can be used as a decision-support tool for financial institutions. The project highlights the potential of machine learning in

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automating and optimizing loan approval processes while reducing operational risk and improving efficiency.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- XGBoost
- SMOTE
- Google Colab

Dataset Source

Kaggle – Loan Approval Prediction Case Study